

$$= 3.28 \text{ N s}$$

**4. Answer: C**

Take the direction to the right as positive.

$$\begin{aligned}\text{Total momentum} &= 5.0 \times 4.0 + 2.0 \times (-3.0) \\ &= 14 \text{ N s}\end{aligned}$$

Total momentum before collision = total momentum after collision

$$14 = (5.0 + 2.0) \times v$$

$$v = 2.0 \text{ m s}^{-1}$$